

Assignment no 05

Title

Convert given binary tree into threaded binary tree. Analyze time and space complexity of the algorithm.

Learning objective.

To understand the basic concept of Non-linear data structure "threaded binary tree" and its use in Data structure.

Learning outcome.

To convert the Binary tree into threaded tree and analyze its time and space complexity.

Theory

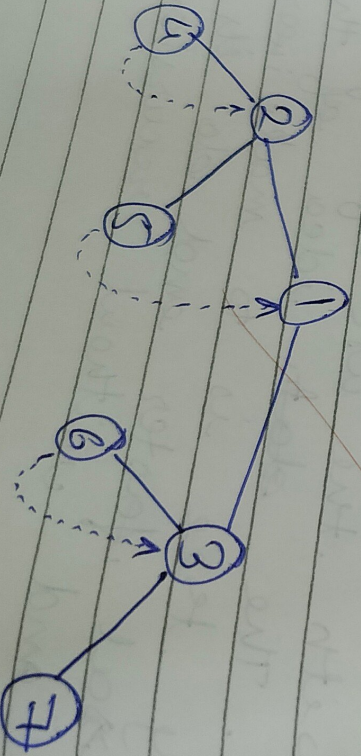
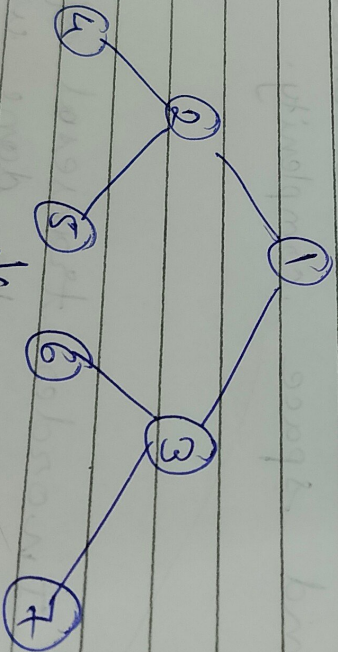
Inorder traversal of a Binary tree is either be done using recursion or with the use of a auxiliary stack. The idea of threaded binary tree is to make inorder traversal faster and do it without stack and without recursion.

There are two type of threaded tree.

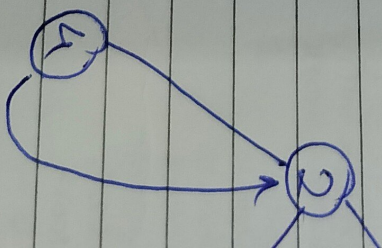
Single threaded when a null right pointer made to point to the in order successor.

Double threaded

where both left and null pointer are made to point to in order predecessor and in order successor respectively.



single threaded structure

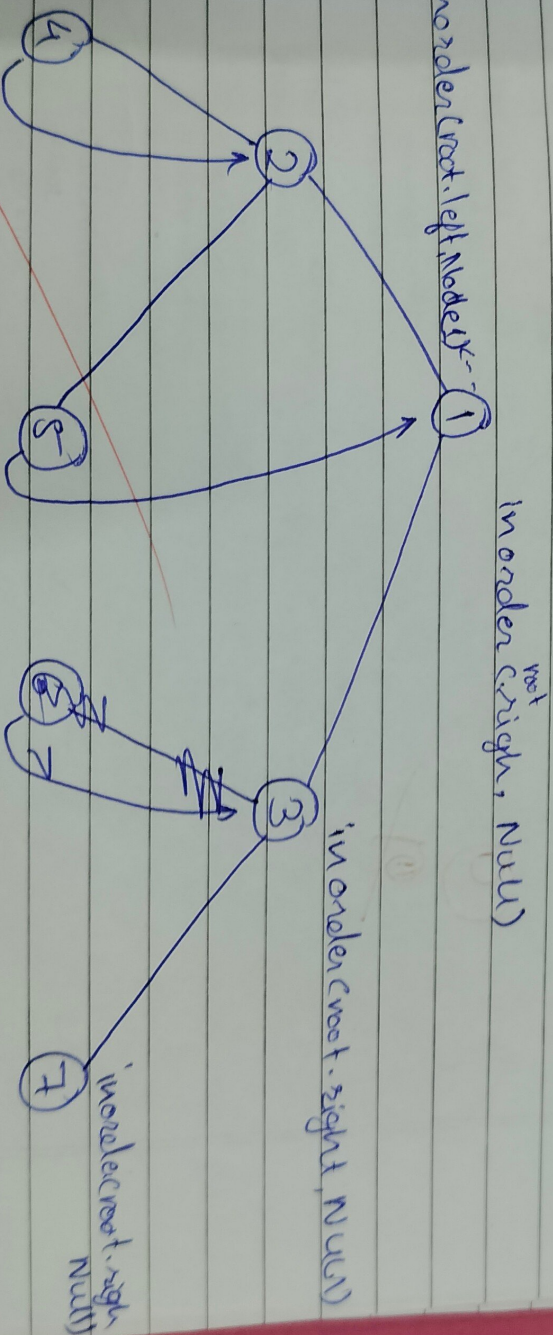


in order right child provide

needed C representation of a threaded Node.
Following is C representation of a single threaded node.

struct Node

{
int data;
Node * left, * right;
bool right Thread;
};



inordercreat.left, Node) inordercreat.left, Node)
right child is null right child is null
provide thread provide the thread

Conclusion

Able to convert the Binary Tree threaded binary tree and its time space complexity.

①

②

periment NO
C++ program
Thread
include <bits/
ing namespa

Structure of
struct Node

int key
Node

// Use
// is a
// to
bool

Converts t
binary tree
This functi
root.
de *create

// Ba
//
if (ro

if (ro

// Fi
if (ro
{

Assignment No 7.

Title

You have a business with several offices; you want to lease phone line to connect them up with each other, and the phone company charges different amount of money to connect different pairs of cities.

Learning objective

- To understand the concept and basic of spanning.
- To understand Graph in Data structure.

Learning outcome

- To implement the spanning
- To find the minimum distance between the vertices of graph in Data structure.

Theory

Properties of greedy Algorithm.

- At each step the best possible choice is taken and after that only the

sub-problem is solved.
sub-problem algorithm might be greedy choices but it can be many choices upon sub-problems depending on and neither

Minimum Spanning Tree

A minimum spanning graph is a kind of sub-graph of undirected graph in which all the graph spans an includes all the has a minimum total edge

Properties of Prim's Algorithm

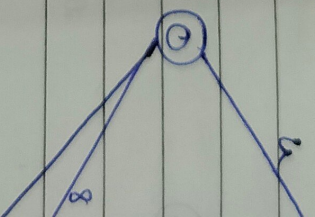
1. The edge in the sub-graph some minimum spanning tree from a single tree.

2. It grows the tree until all the vertices of the graph

3. An edge is added a cut at each every step of any if its weight is connecting it to a vertex of

Algorithm

1. Begin with
2. Find a vertex in is not in
3. Stop and

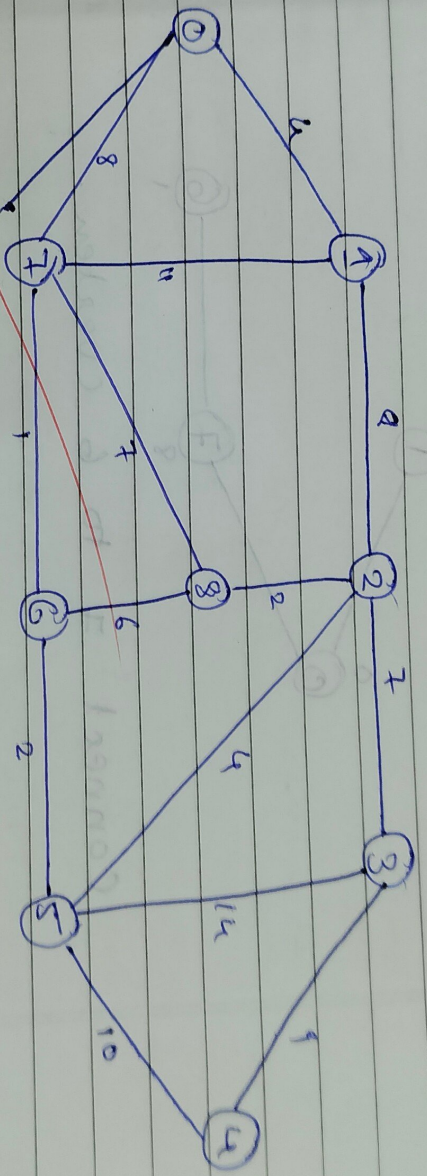


start.

connect

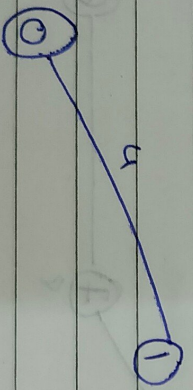
Algorithm.

1. Begin with any vertex which you think.
2. And an edge that connected any vertex in the tree to any vertex that is not in the tree.
3. Stop when $n-1$ edge.

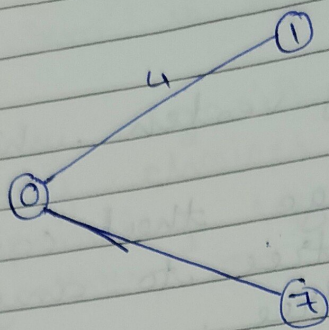


Example.

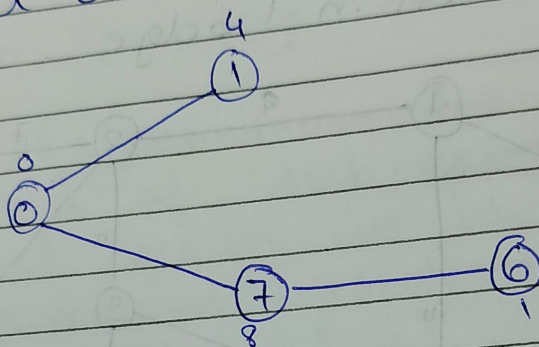
1. Start.



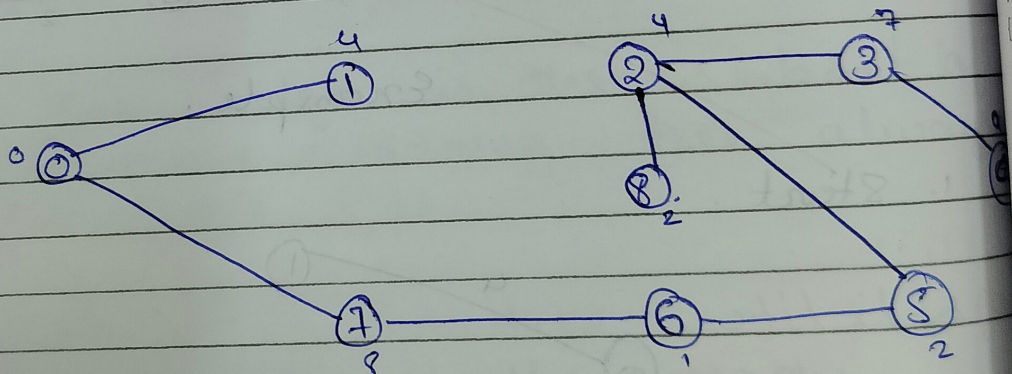
connect 0 to 1



connect 0 to 7 (vertices).



Connect 7 to 6 (vertices).



Conclusion

Understand the concept of basic of spanning and to find the minimum distance between the vertices.

Assignment No - 12 .

Title

Company maintain employee info as employee ID name designation and salary . Allow user to add , delete info of employee .

Learning objective

- To understand concept of file organization in data structure
- To understand concept & feature of Indexed Sequential file organization

Learning outcome .

- Define class for sequential file using object oriented feature .
- Analyze working of various operations on Indexed Sequential Access Method .

Theory File organization refers to the relationship of the key of the record .

to the physical location of the record in the computer file.

File organization may be either physical file or a logical file. A physical file is a physical unit, such as magnetic tape or a disk. A logical file on the other hand is a complete set of records for a specific application or purpose.

Some of the file organization are

1. Sequential file organization
2. Hash / Direct file organization.
3. Indexed / Sequential Access Method
4. B+ tree file organization.
5. Cluster file organization.

Index

sequential
method
using

Do

R1
R2
R5
R7
R8

R9

Pros

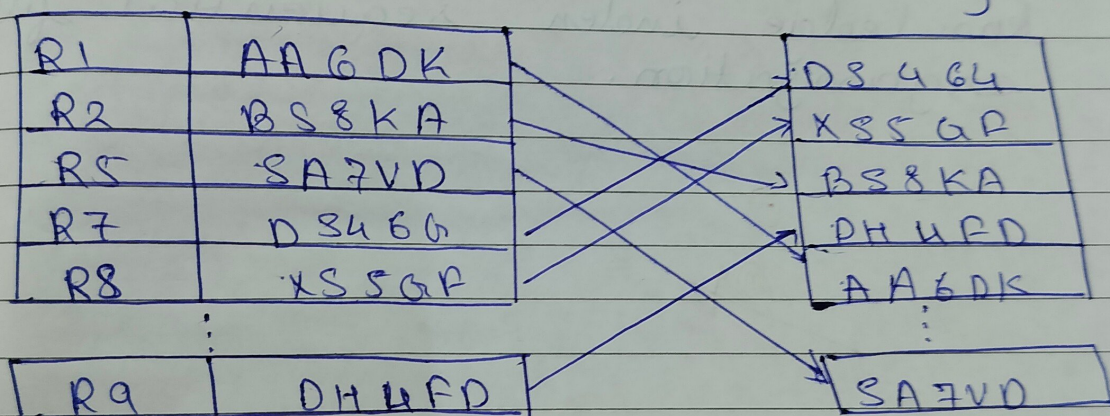
has
search
is

Indexed sequential access method (ISAM)

ISAM method is a advanced sequential file organization. In this method record are stored in the file using the primary key.

Data Record

Data Block in memory



Pros of ISAM.

In this Method each record has the address of its cluster block. Search a record is a huge database is quick and easy.

Cons of I-SAM.

- This method requires extra space in the disk to store index value.

Conclusion

This program gives us the knowledge index sequential file organization.

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